



Calibration of multi-spectral photosynthetically active radiation sensor

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ABSTRACT

Photosynthetically active radiation (PAR) sensors are widely used in plant stress monitoring, ecophysiology, and forest growth modeling. Recently, multi-spectral PAR sensors have emerged as practical alternatives to established single-channel systems. These sensors offer significant potential, given their competitive accuracy to commercial systems and additional capability to obtain spectral information. This work introduces a new calibration framework specifically tailored for multi-spectral sensors. Previous approaches were only accurate and reliable under identical light conditions. Our calibration setup within a controlled environment ensures consistency and reproducibility of the sensor calibration in every ambient light condition. A dedicated data acquisition and preprocessing framework accounts for spectral characteristics of each component, forming the basis for the Partial Least Squares (PLS) calibration model. This fits the multi-spectral data to the ideal quantum response and enables robust modeling even in case of high collinearity among input variables. With only one calibration coefficient for each channel, a highly efficient implementation of the PAR calculation is possible. Moreover, the method compensates for the spectral leakage of the NIR in the visual range channels. The spectral accuracy of the calibration model is demonstrated by presenting, for the first time, the quantum response curve of a calibrated multi-spectral PAR sensor. In addition, a field validation experiment is conducted under variable weather conditions to assess the absolute performance. With an error of 2.43% relative to a commercial sensor, this work highlights the accuracy achieved with the proposed calibration method. For further investigation, this work provides a repository with data from our calibration setup.

1. Introduction

Measurement of photosynthetically active radiation (PAR) is essential in a wide range of research areas including plant stress monitoring, ecophysiology, and forest growth modeling [1]. PAR is the main source of energy for photosynthetic processes and is defined as the photon flux density in the range of 400 nm to 700 nm [2,3]. PAR functions as a fundamental parameter for understanding light-dependent biological and ecological processes. By monitoring PAR, scientists can predict how changes in light availability impact plant and ecosystem productivity. In recent years, multi-spectral PAR sensors have been introduced by researchers as practical alternatives to commercial systems [4–7]. To take full advantage of these sensors, a calibration method is needed that operates independently of ambient light conditions and ensures consistent and reliable sensor performance. In this work, a method is proposed to simultaneously characterize the sensors spectrally and calculate an absolute calibration coefficient for each channel of the multi-spectral sensor. This method could also help researchers in the fields of colorimetry and spectroscopy, as well as health monitoring, where the qualitative and quantitative dimensions of light are required [8–10].

For the field of PAR sensing, this method will provide the quantum response curve. In addition, demonstrating the high accuracy in a field validation experiment, this calibration represents a further step toward a new class of PAR sensors that not only have competitive accuracy compared to commercial systems, but also provide spectral information about the incident light [7,11]. Due to their compact size, low power requirement, and intrinsic spectral channels which eliminate the need for extra filters, multi-spectral sensors are ideal for large-scale energy-autonomous sensor networks. This feature is utilized in a sensor network developed at the ECOSENSE collaborative research center of the University of Freiburg [12]. ECOSENSE thus uses a remote naturally grown forest as a living laboratory to evaluate and explore strongly miniaturized and energy-efficient sensors. Here, PAR is utilized in various studies to assess stress effects in plants: (a) with incident PAR, the quantum use efficiency of the photosynthetic apparatus is evaluated in chlorophyll fluorescence measurements. (b) Similarly, to separate the effects of light in photosynthetic activity under stressed and non-stressed conditions, it is crucial to assess the PAR. For investigations (a) and (b), it is necessary to attach the sensor directly to the leaf

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for several months. This demands a compact and lightweight setup to ensure that the leaf's natural behavior remains unaffected, an ideal use case for multi-spectral PAR sensors.

Conventionally established methods for measuring PAR are spectroradiometers and photodiodes. None of them, however, embodies the advantages of multi-spectral sensors [13,14]: (1) A spectroradiometer is the most accurate device for measuring PAR. It typically includes a diffraction optic and a photodiode array that provide high spectral resolution in the nanometer range. These devices are used for measurements of PAR, where additional spectral information is required. However, for most long-term measurement campaigns, a spectrally high resolution of PAR is not required, and spectroradiometers are in terms of size and initial cost not competitive with other solutions. (2) Photodiodes are the second method of measuring PAR. Providing accuracy and size at an affordable price, they are considered state of the art for measuring long periods in environmental science [13]. All photodiode-based sensors are constructed in a similar manner: a band pass filter for the PAR spectrum from 400 nm to 700 nm, a broadband photodiode, and a special weighting filter [1,14]. The intensity with which optical detectors respond to the energy of a single photon is known as the quantum response. The energy of a single photon is directly proportional to its wavelength. To assess PAR effectively, the photodiode's photon response should be uniform across all wavelengths within the specified range. Producers of single-channel PAR sensors employ advanced filter technology to adjust the quantum response of the photodiode, guaranteeing a uniform photon response regardless of wavelength. However, this filter technology results in single channel PAR sensors that lack the ability to provide information about the spectral composition. To combine the advantages of a spectroradiometer with photodiode-based PAR sensors, researchers introduced multi-spectral photodiode sensors [4,6,7,15]. These kinds of sensors are off-the-shelf devices and incorporate an array of narrow band photodiodes, representing channels. Each channel of these sensors features high-precision optical interference filters integrated into a silicon photodiode to enable the measurement of specific spectrum segments [16].

For laboratory calibration of photodiode-based PAR sensors, typically a standard quartz tungsten halogen lamp with known spectral irradiance is used to determine the absolute spectral response and calculate the calibration coefficient for the PAR sensor [1,17]. If a laboratory-calibrated PAR sensor is already available, several single-channel PAR sensors can be calibrated at the same time under solar radiation in conjunction with the calibrated PAR sensor used as a reference sensor [1]. In cases where the spectral sensitivity of all sensors within the calibration under solar radiation is identical, corrections are unnecessary. In contrast, when sensors showing varied spectral sensitivities are employed, precise calibration requires the use of the spectral response curve to compensate for deviations [13]. For the calibration of multi-spectral PAR sensors, researchers have followed until now the approach using a calibrated PAR sensor under solar radiation as a reference. Early results were achieved by adjusting the output of the four channels of an RGBW sensor to a commercial photodiode-based PAR sensor and solar radiation in order to calibrate the sensor output for PAR [4]. Likewise, a multi-spectral sensor with eight channels in the PAR range was fit to the output of a commercial photodiode-based PAR sensor with data from open-sky measurements to calculate calibration coefficients for each channel [6,15]. This approach is limited, since the spectral response of the multi-spectral sensor and the single-channel photodiode sensor used as reference, as well as the spectral composition of the light source are not considered [13,17]. In the mentioned publications, a spectral correction is not done between different sensor systems, resulting in calibration coefficients that are only correct for the specific light spectrum during the calibration process.

For the computation of the calibration coefficients, in previous works a multi-linear regression (MLR) scheme is employed to align

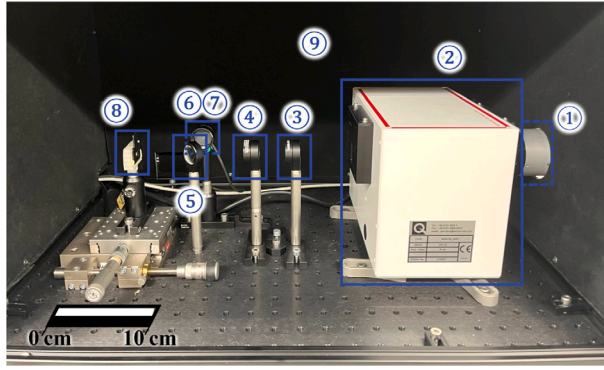
the response of multi-spectral sensors with a commercial photodiode-based PAR sensor [4,6]. Another method used for the calibration of multi-spectral sensors is the partial least squares (PLS) regression [15]. PLS is mentioned in spectroscopy as an effective approach also in the case of correlated channels and of multivariate calibration of chemical concentrations [18,19]. In a different approach, instead of one, three multi-spectral sensors are used to span an extensive spectral range within the visible (VIS) and NIR bands, including a total of 11 channels relevant to the PAR spectrum [7]. Here, a spectroradiometer is utilized as a calibration standard, instead of a photodiode-based PAR sensor. Due to the spectroradiometer, the spectral response of each individual sensor as well as the spectral composition of a light source are known. The calculation of the calibration coefficients is done by linearly mapping the multi-spectral sensor response of each channel to the spectroradiometer output of the same waveband. The data of the multi-spectral sensor was calibrated using a vector quantization with k-means clustering. The algorithm was trained on several light sources and lighting conditions. The derived correction factors were applied to new measurements to correct for spectral leakage in the NIR, reconstruct the corrected spectrum, and interpolate missing bands. This approach enables differentiations between various light compositions and demonstrates a 6 % higher accuracy of multi-spectral sensors compared to a commercial photodiode-based PAR sensor under different light sources indoors and outdoors. However, the study lacks a controlled calibration light regime, making the method unsuitable for calibrating a substantial number of sensors as required by the autonomous sensor network in ECOSENSE. Moreover, the k-means correction algorithm that addresses spectral leakage requires training across multiple light regimes.

Prior deficiencies are addressed in this work through the establishment of a controlled environment. The customized setup enables the calibration of each sensor under consistent conditions. Moreover, this work presents a data acquisition and preprocessing approach that accounts for the spectral characteristics of each device, ensuring that the absolute response at every wavelength for each channel within the spectrum of interest is measured. Subsequently, the PLS algorithm has been selected for calibration due to its effectiveness in handling highly correlated data [15]. In this work, the model described in [15] serves as the foundation and has been further optimized to account for spectral leakage in the near-infrared (NIR) range. The PLS algorithm produces a single coefficient corresponding to each channel, enabling efficient PAR computation. This study validates the calibration by presenting, for the first time, the quantum response curve of a calibrated multi-spectral PAR sensor. In contrast to previous works, this calibration to the quantum response curve ensures a spectrum-independent response. It also demonstrates the sensors performance in a field validation experiment under varying weather conditions. The high performance under different light conditions highlights its competitiveness with commercial sensors.

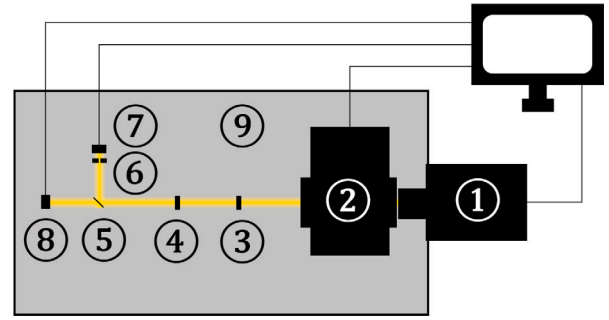
2. Materials and methods

2.1. Photosynthetic photon flux density

The spectral irradiance, or radiant flux density $I^E(\lambda)$, serves as the fundamental parameter in radiometry [17]. Characterizes the radiant flux impacting a specified area within an infinitesimally narrow spectral range. The measurement of the radiant flux is expressed in W , therefore, spectral irradiance is quantified in $W\ m^{-2}\ nm^{-1}$. Since photosynthesis is a quantum process, typically, incident light is represented as the number of photons in moles using Avogadro's constant [13]. Consequently, the quantum unit for spectral irradiance is quantified in $\mu mol\ photons\ m^{-2}s^{-1}nm^{-1}$ and is called spectral photon flux density (PFD). In accordance with the conventions in the literature, in this work, radiant quantities expressed in energy units are denoted by the superscript index E, while those expressed in quantum units are



(a) Image of the measurement setup inside the box.



(b) Diagram of the measurement setup.

Fig. 1. (1) lamp, (2) monochromator, (3) engineered diffuser, (4) collimator, (5) beam splitter, (6) 5 mm Aperture, (7) reference sensor, (8) DUT and (9) light isolated box.

indicated by the index Q [17]. This relationship can be articulated as follows:

$$I^Q(\lambda) = \frac{\lambda I^E(\lambda)}{N_A h c} = \frac{\lambda I^E(\lambda)}{\gamma} \quad (1)$$

where h is Planck's constant, c is velocity of light, λ is wavelength in nm, N_A is Avogadro's quantity and $\gamma = N_A h c = 119.8 \text{ W s nm } \mu\text{mol}^{-1}$ the resulting conversion factor. Transforming equation (1) to the integral in the PAR range, it is known as the photosynthetic photon flux density (PPFD) [2]:

$$I_{\text{PPFD}}^Q = \frac{1}{\gamma} \int_{400 \text{ nm}}^{700 \text{ nm}} \lambda I^E(\lambda) d\lambda \quad (2)$$

2.2. Measurement setup

The calibration measurements were carried out in a light-insulating box in Fig. 1(a) to ensure that the outside light did not interfere with the sensor. To eliminate reflections, a black flocked paper was applied to the side walls of the isolated box. The configuration shown in Fig. 1(b) includes the following components: (1) the light source, a Quartz Tungsten Halogen (QTH) lamp, (2) a Czerny-Turner monochromator MSH 150, (3) an ED1-S20 engineered diffuser, (4) a collimator LA1134-A-ML and (5) a beam splitter BSW26. As measurement devices, there are (7) the reference sensor S120C with (6) 5 mm diameter aperture, and (8) the device under test (DUT) to be calibrated. The QTH lamp was chosen for its stable intensity in the desired spectral range. The monochromator in series with the lamp enables the irradiation of the sensor with monochromatic light in a bandwidth of 5 nm. The engineered diffuser and collimator reshape the wave front, collect divergent light, and increase the intensity. The beam splitter separates light into two beams, one toward the reference sensor and one to the DUT. The reference sensor takes over the function of the calibration standard. (Components 1 and 2: Quantum Design, Pfungstadt, Germany, components 3–5 and 7: Thorlabs, Newton, New Jersey, United States).

2.3. Characterization of the setup

In preparation for the primary measurements, a comprehensive characterization of all the components of the experimental setup was performed to validate the configuration and ultimately ensure precision.

2.3.1. Beam profile

The monochromator setup emits light with a nearly Gaussian wavefront, characterized by a higher intensity in its center compared to the edges. As the detection area of the multi-spectral sensor and the reference sensor is different, it is crucial that the optical power per area, the

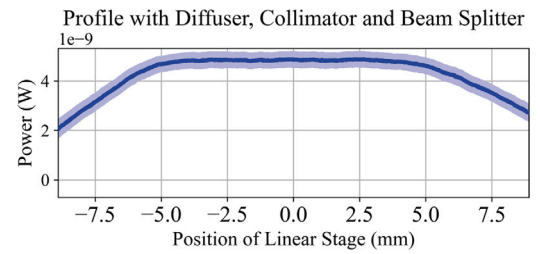


Fig. 2. The beam profile from the monochromator, which has been altered by the engineered diffuser, produces a 10 mm in diameter flat-top beam. This ensures equal irradiance on both the reference sensor and the DUT.

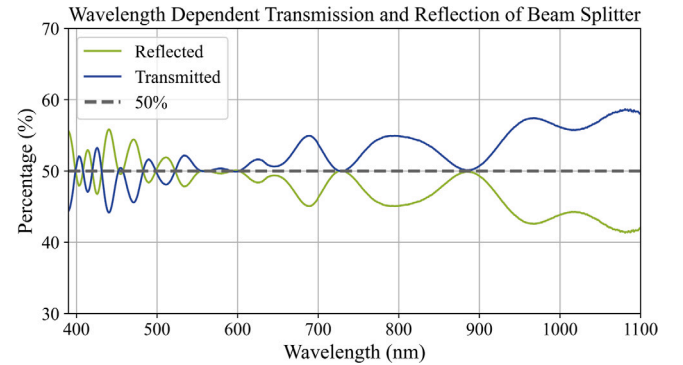


Fig. 3. Transmission and Reflection characteristics of beam splitter, by wavelength.

irradiance, is identical for both devices. Therefore, the light output of the monochromator needed to be reshaped to achieve a uniform spatial light distribution to become independent of the detection area. Fig. 2 shows the obtained pseudo-flat top beam profile with 10 mm diameter created by the engineered diffuser. Using apertures smaller than 10 mm on both the reference sensor and the DUT, uniform irradiance can be assumed in both detection areas.

2.3.2. 50:50 Beam splitter

A beam splitter is utilized to synchronously direct incident light to both the reference sensor and the multi-spectral sensor for measurement. The dielectric coating of the beam splitter causes wavelength-dependent optical properties, which must be taken into account for calibration. To address the variations in transmission and reflection with frequency, as shown in Fig. 3, the reference sensor was used to

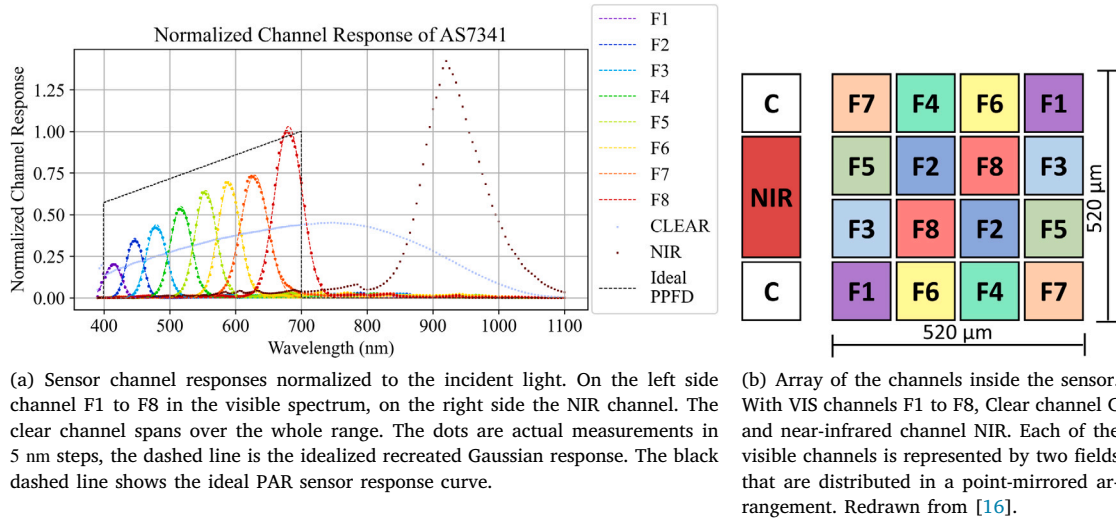


Fig. 4. Spectral responses of the channels and the configuration of the channels on the sensor chip of the AS7341 multi-spectral sensor.

measure them. Based on these measurements, the transfer function was obtained by sweeping the monochromator with increments of 1 nm from 390 nm to 1100 nm. For each wavelength the specific ratio $\kappa_{BS}(\lambda)$ was computed:

$$\kappa_{BS}(\lambda) = \frac{P_{transmitted}^E(\lambda)}{P_{reflected}^E(\lambda)} \quad (3)$$

2.4. Multi-spectral sensor

Multi-spectral sensor systems consist of arrays of silicon photodiodes. Each position within the array acts as an individual photodiode channel with a bandpass filter. Each channel is defined by a distinct full width at half-maximum (FWHM) and peak wavelength λ_p . Fig. 4(a) illustrates the responses of the channels and Fig. 4(b) the channel configuration on the sensor chip of the AS7341 multi-spectral sensor (ams-OSRAM AG, Premstaetten, Austria). This sensor will serve as the DUT for this study, as it was already used successfully to measure PAR in three other studies [6,11,15].

As shown in Fig. 4(a), this device has 8 channels in the VIS range, one NIR channel, and one clear channel with sensitivity throughout the silicon response [16]. The 4×4 sensor matrix, shown in Fig. 4(b), contains two fields per channel. The whole sensor area measures $520 \mu\text{m} \times 520 \mu\text{m}$. The clear channel and the NIR channel are located next to the array with the VIS channels. Due to the matrix arrangement, the channel's response to light from different directions is different. For optimal sensor utilization, the OptSaver L-52 diffuser (Kimoto LTD, Cedartown, GA, USA) is therefore used to ensure an almost cosine-shaped response and to illuminate the channels evenly [20]. The AS7341 exhibits spectral overlaps with neighboring channels, experiences some degree of crosstalk, and has leakage through the bandpass filter at non-target wavelengths. Consequently, these systems require software-based post-processing.

The sensor is equipped with analog-to-digital converters (ADC) that offer a high linear dynamic input range by adjusting the conversion gain in 11 steps from $0.5\times$ to $512\times$ and the integration time up to 47 s. The linearity of the sensor in different ADC settings is given by the manufacturer and is also proven in the supplementary material of this work [21]. To make every measurement independent from integration time and gain, it has to be scaled like in Eq. (4):

$$s_{i,basic} = \frac{\text{raw}_i}{g \times \text{tint}} \quad (4)$$

where raw_i is the raw sensor value of channel i , g is the gain and tint the integration time of the ADC. The value $s_{i,basic}$ is the independent

value of the channel i . The basic value allows the comparison of measurements performed under different ADC settings for varying light intensities. In order to attenuate the typical temperature-dependent effects, the AS7341 has an integrated dark photodiode with an integrated auto-zeroing function. A demonstration of the temperature-independent response of the multi-spectral sensor can be found in the additional material [21]. To enable the multi-spectral sensor to measure PAR the sum of all channels should follow the ideal quantum response curve, represented by the black curve in Fig. 4(a). For that purpose, each channel needs to be weighted separately, therefore, the sensor needs to be calibrated.

2.5. Calibration

2.5.1. Data acquisition

For the acquisition of calibration data, the monochromator was utilized to illuminate both the reference sensor and the DUT, with increments of 3 nm ranging from 390 nm to 700 nm and increments of 5 nm from 700 nm to 1,100 nm to cover the PAR spectrum and the NIR of the AS7341. The sweep was repeated five times to obtain a number of five times 185 monochromatic measurements for training the model used for the calibration. Under the assumption of a linear relationship of the AS7341 with the irradiance in Section 2.4, the measurements were repeated only at one distinct power level of the lamp. Each increment results in a single measurement from the reference sensor, ten data points, one per channel, are captured by the spectral sensor, a single assessment of both sensor gain and integration time, and the distinct wavelength provided by the monochromator.

2.5.2. Preprocessing

In the first step, under the assumption of a flat top beam, the spectral irradiance $I_{ref}^E(\lambda)$ of the incoming light on the reference sensor is determined by using the specified area A_{ref} of the aperture that covers the reference sensor with a diameter of 5 mm, in conjunction with the spectral power $P_{ref}^E(\lambda)$ recorded at the reference sensor:

$$I_{ref}^E(\lambda) = \frac{P_{ref}^E(\lambda)}{A_{ref}} \quad (5)$$

The spectral irradiance directed at the DUT, denoted as $I_{DUT}^E(\lambda)$, is derived using $I_{ref}^E(\lambda)$ from Eq. (5) present at the reference sensor and the wavelength-dependent transmission coefficient $\kappa_{BS}(\lambda)$ from the beam splitter as described in Eq. (3):

$$I_{DUT}^E(\lambda) = I_{ref}^E(\lambda) \cdot \kappa_{BS}(\lambda) \quad (6)$$

Given the inherent properties of the QTH lamp, $I_{\text{DUT}}^E(\lambda)$ does not exhibit uniform intensity in the spectral range. Therefore, it is necessary to normalize the raw data of each channel of the spectral sensor $s_{i,\text{basic}}(\lambda)$ by $I_{\text{DUT}}^E(\lambda)$:

$$s_{i,\text{norm}}(\lambda) = \frac{s_{i,\text{basic}}(\lambda)}{I_{\text{DUT}}^E(\lambda)} \quad (7)$$

As well as to normalize the $I_{\text{DUT}}^E(\lambda)$ by itself:

$$I_{\text{DUT, norm}}^E(\lambda) = \frac{I_{\text{DUT}}^E(\lambda)}{I_{\text{DUT}}^E(\lambda)} = 1 \quad (8)$$

The $s_{i,\text{norm}}(\lambda)$ values of all visible channels F1 to F8 and the Clear and NIR channels form the input matrix for the subsequent calibration procedure. The normalized spectral irradiance that falls on the spectral sensor $I_{\text{DUT, norm}}^E(\lambda)$ is converted with Eq. (1) to its quantum unit, denoted as $I_{\text{DUT, norm}}^Q(\lambda)$. This forms the target value for the subsequent calibration procedure. To target the PAR range, for all wavelengths below 400 nm and above 700 nm, $I_{\text{DUT, norm}}^Q(\lambda)$ is set to 0.

2.5.3. PLS calibration model

For the computation of the calibration coefficients, the Partial Least Squared (PLS) algorithm is utilized. Previous research has shown good performance of PLS with multi-spectral sensor overlapping channels [15]. PLS is a multivariate regression method based on the same principles as Principal Component Analysis (PCA) and Principal Component Regression (PCR). PCA reduces data dimensionality by transforming data into new directions, called principal components (PC), which capture the most variance with minimal information loss [22]. Similarly, PLS transforms the input data into principal components that maximize both data variance and correlation with the target variable.

The dataset was split in training and validation data. All monochromatic increments of four of the five sweeps were used to train the model, and one sweep to validate it. Compared to the research conducted in [15], our model not only incorporates channels F1 through F8, but also includes the Clear and NIR channels to account for spectral leakage of some channels in the NIR range. The input matrix X , consisting of $s_{i,\text{norm}}(\lambda)$, contains the preprocessed readings of the ten channels. The target variable Y corresponding to $I_{\text{DUT, norm}}^Q(\lambda)$, represents the normalized calculated PAR as described in Section 2.5.2. The PLS algorithm is applied in three steps:

1. First, the hyperparameter selection step to find the optimal number of PCs. The number of PCs, ranging from 1 to 10, is determined by evaluating the performance of the model through a 10-fold cross-validation using the training set. For each number of components, a PLS regression model is trained and its performance is assessed using R^2 . The number of components that produce the highest R^2 depends on the DUT. In this study, between 6 and 8 PCs were observed.
2. Second, the final model training. The number of components that produce the highest R^2 is selected. The final PLS model is then trained using this optimal number of PCs on the training set, and the regression coefficients C_i are computed for each input feature.
3. Third, the validation of the training. To evaluate the performance of the model, the coefficients obtained are applied to the validation data. The performance is assessed using R^2 .

For this work, an accompanying repository with the code and data utilized was published to allow reproduction and enhancement of the results [21].

2.6. PAR computation

The implementation of PLS algorithm provides a coefficient C_i for each input variable. In Eq. (9), these coefficients are multiplicative

factors that enable the transformation from unit-less digital sensor readings to PAR. Eq. (9) is the discrete version of Eq. (2) for multi-spectral sensors. To validate the performance of the resulting calibration coefficients, they were applied to the validation data to see if the model follows the ideal quantum response. For multi-spectral sensors, the I_{PPFD}^Q is calculated from the weighted sum of all basic channel responses $s_{i,\text{basic}}$:

$$I_{\text{PPFD}}^Q \approx \sum_{i=1}^n C_i s_{i,\text{basic}} \quad (9)$$

2.7. Evaluation of uncertainty

The measurement uncertainty of the absolute measurements is composed of the given uncertainty of the reference sensor S120C and the repeatability of the DUT (see Eq. (11)). The uncertainty of the reference sensor is given by $\pm 5\%$ for the spectral range of 400 nm to 439 nm and $\pm 3\%$ for the spectral range of 440 nm to 980 nm. The relative repeatability uncertainty u_{repDUT} of the DUT is computed for each wavelength based on the standard deviation divided by the mean across the five sweeps:

$$u_{\text{repDUT}}(\lambda) = \frac{\sigma_{\text{DUT}}(\lambda)}{\bar{x}_{\text{DUT}}(\lambda)} \cdot 100\% \quad (10)$$

with $\sigma_{\text{DUT}}(\lambda)$ the standard deviation and $\bar{x}_{\text{DUT}}(\lambda)$ the mean of the calculated PAR. The measurement uncertainty for the absolute calibration is calculated for four different wavelength ranges to consider the reference sensor specification and the PAR spectrum: below 400 nm, above 700 nm, from 400 nm to 439 nm and from 440 nm to 700 nm. u_{repDUT} in Eq. (11) is the arithmetic mean of all $u_{\text{repDUT}}(\lambda)$ in the specific wavelength range:

$$u_{\text{abs}} = \sqrt{u_{\text{ref}}^2 + u_{\text{repDUT}}^2} \quad (11)$$

with u_{abs} the uncertainty of the absolute measurement, u_{ref} the uncertainty of the reference sensor and u_{repDUT} the relative uncertainty of the repeatability.

2.8. Field validation experiment

To validate the calibration of the multi-spectral PAR sensor, a field validation experiment was conducted from March 26 to 28, 2025, on the campus of the University of Freiburg (coordinates: 48.014579/7.830953). One calibrated multi-spectral PAR sensor was installed adjacent to a reference sensor to enable direct comparison. As reference, a commercially calibrated Onset S-LIA-M003 PAR sensor was used. The multi-spectral sensor recorded measurements at 30-second intervals, while the reference sensor sampled every 10 s. The solar spectrum, often described in terms of its color temperature, varies throughout the day and under different atmospheric conditions [23]. To adequately capture these variations and prove spectral independence, measurements must span complete diurnal cycles and a range of weather scenarios. On 26 March, the sky was predominantly overcast, resulting in diffuse light conditions with minimal direct solar radiation. March 27 was characterized by rapidly changing conditions, with intermittent sunshine and frequent cloud cover, leading to highly variable irradiance levels. Such conditions may introduce minor mismatches between the datasets because of their differing temporal resolutions. On 28 March, the weather was mostly clear, providing stable and consistent radiation measurements throughout the day.

3. Results and discussion

The effectiveness of the calibration approach is assessed through comparison with the ideal quantum response curve and a field validation experiment, highlighting improvements over previous methods. Key aspects evaluated include the uncertainty of repeatability across the five sweeps, variability between the four DUTs, and the role of NIR compensation in reducing spectral crosstalk and enhancing calibration robustness.

Table 1

Uncertainty measurements across different wavelength ranges.

Wavelength range (nm)	Reference sensor (%)	Relative uncertainty of repeatability (%)	Measurement uncertainty (%)
390–399	5	1.253 ± 0.460	5.155
400–439	5	0.232 ± 0.015	5.005
440–700	3	0.102 ± 0.018	3.002
701–1100	3	3.182 ± 1.185	4.373

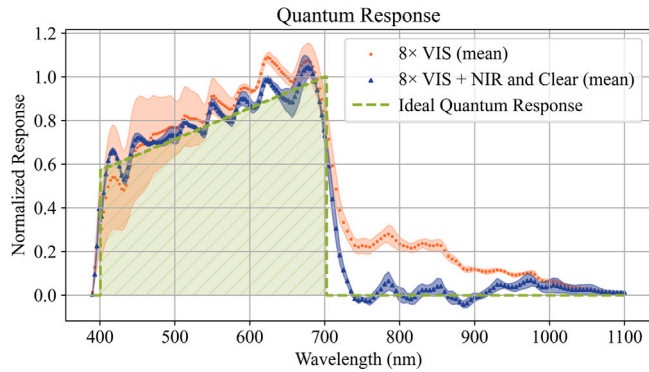


Fig. 5. Mean with SD of the quantum response of the four sensors compared to the ideal quantum response curve. With an R^2 of 94.788 % (SD ± 0.859 %), this curve exhibits a strong correspondence to the ideal curve.

3.1. Uncertainty of repeatability

The values of the four different wavelength ranges in Table 1 are arithmetic means for the four sensors used. In the PAR range of 400 nm to 700 nm the relative uncertainty of repeatability u_{repDUT} shows a low error of 0.232 % and 0.102 %, which has a negligible influence on the uncertainty of absolute measurements u_{abs} . Outside the PAR range, repeatability is worse and leads to higher measurement uncertainties. The wavelength reproducibility of the monochromator as well as the lamp stability can be neglected for the uncertainty calculation. Therefore, these higher repeatability uncertainties are probably due to the sensor itself, as a result of its lower spectral resolution outside the PAR range. To further decrease the measurement uncertainty, a more accurate reference sensor is necessary.

3.2. Uncertainty between sensors

The calibration coefficients obtained during the PLS calibration process are applied to the unitless digital values for each channel at each wavelength step and summed according Eq. (9). The data effectively reflect the quantum response of the multi-spectral sensor, as depicted in Fig. 5. To investigate the variability between the four sensors, the arithmetic mean of the calculated PAR for each wavelength is shown in blue triangles. The blue shaded area around the curve shows the standard deviation. The ideal PAR is the dashed green line. The accuracy of the calibration is expressed as the R^2 between the ideal PAR and the calculated PAR. With an arithmetic mean of $R^2 = 94.788$ % and a standard deviation of ± 0.859 % the variability between the sensors is very low.

3.3. NIR compensation

Furthermore, the calibration model used in this work introduced an efficient NIR compensation. Since the AS7341 sensor filters are not ideal, some channels in the blue and green regions exhibit spectral leakage in the NIR range. The orange curve in Fig. 5 was obtained by using the eight visible channels in the calibration algorithm [6, 15]. As we observe a response outside the 400–700 nm range, which

should be 0 for PAR, we can therefore conclude that the response is affected from spectral leakage. With an $R^2 = 79.858$ % (± 3.448 % SD) compared to the ideal response, this approach is unsuitable for accurate PAR measurements. By incorporating the Clear and NIR channels into the PLS calibration model, we are able to correct for spectral leakage. The blue curve in Fig. 5 shows the quantum response with NIR compensation. The fit of the blue curve in Fig. 5 to the ideal PAR showed a significantly higher coefficient of determination, with $R^2 = 94.788$ % (± 0.859 % SD). This indicates that the inclusion of the Clear and NIR channels effectively compensates for crosstalk from the visible channels in the near-infrared region, thereby improving the fit of the sensor response with the ideal quantum response curve. Previous calibration approaches such as those in Table 2 accounted for the crosstalk in the NIR region implicitly within the calibration procedure under solar radiation. The limitation of such approaches lies in their dependence on the spectral composition of the incident light during calibration, making the resulting calibration valid only under similar spectral conditions. To illustrate this limitation, consider an extreme case. If calibration is performed under a light regime dominated by blue light, only the sensor channels F1 and F2, which are sensitive to blue wavelengths, will receive meaningful calibration coefficients. The coefficients for other channels will remain near zero. If the sensor is used subsequently in an environment irradiated primarily by red light, it may produce little or no PAR signal despite the radiation being inside the valid PAR range. The method presented in this work addresses this limitation by implementing an absolute calibration of the quantum response curve. As a result, the calibrated PAR measurements become independent of the spectral distribution of the incident light in the measurement environment.

3.4. Field validation experiment

The field validation experiment under natural environmental conditions demonstrated that the multi-spectral PAR sensor performed comparable to that of the calibrated commercial reference sensor. Fig. 6 presents representative data from two contrasting days: the overcast day of March 26 and the clear day of March 28. For both days, measurements are shown between 8 am and 8 pm CET. On the intermediate day, March 27, which was characterized by highly variable sky conditions, PAR values of up to $2000 \mu\text{mol}(\text{m}^2 \text{ s})^{-1}$ were reliably measured. Systematic deviations between sensors were observed primarily in the early morning before 10 am and around 6 pm. These discrepancies are attributed to transient shading effects from nearby buildings. To exclude periods affected by transient shadows, only measurements recorded between 11 am and 5 pm were considered for the calculation of error metrics. During this period, a total of 720 measurements were theoretically possible. However, due to time-matching discrepancies between the multi-spectral and reference sensor, only 474 matched measurements from March 26 and 475 matched measurements from March 28 were kept for a solid statistical analysis. On the overcast day, the mean absolute percentage error (MAPE) between the multi-spectral and reference sensor was 7.36 %, while on the clear day the MAPE decreased to 2.43 %.

A comparison of these results with previous multi-spectral sensor calibration methods in Table 2 demonstrates that the presented calibration approach achieves competitive performance. Depending on weather conditions, the calibration showed a lower error or a similar error to those reported in the cited studies.

The high reliability and accuracy of the setup and the calibration method are indicated by the minimal repeatability uncertainty for each wavelength in Table 1 and the low standard deviation of ± 0.859 % between all four sensor calibrations. With an error of 2.43 % during the field validation experiment, the calibration also proved its accuracy outside the laboratory. This study presents a calibration system suitable for multi-spectral PAR sensors independently of environmental

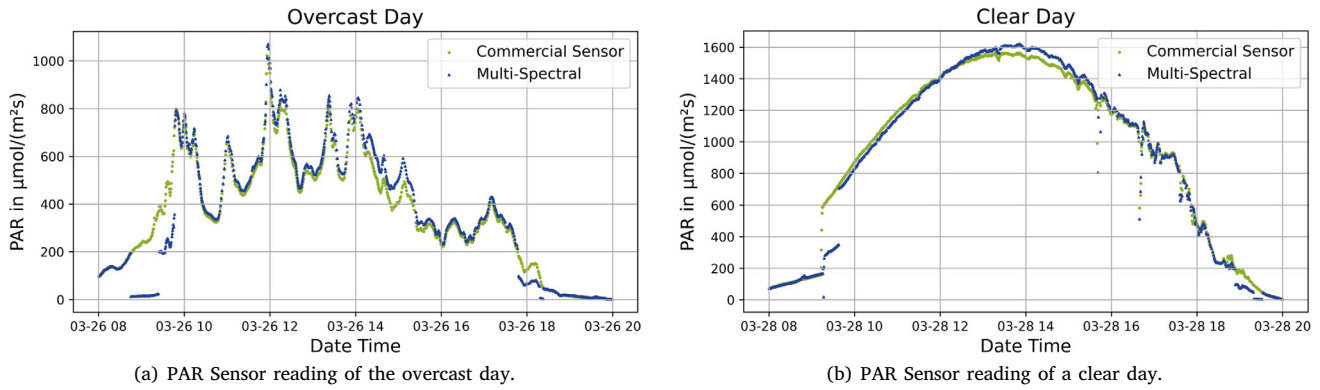


Fig. 6. PAR measurements of the overcast day of March 26 and the clear day of March 28, 2025. As reference, a commercially calibrated Onset S-LIA-M003 PAR sensor was used.

Table 2
A comparison of the field test results with previous multi-spectral sensor calibration methods.

Work	Calibration reference	Calibration conditions	R ² (%)	MAPE (%)	nRMSE (%)
This work (clear)	Photodiode power meter	Controlled monochromatic light	93.91	2.43	4.01
This work (overcast)			94.13	7.36	5.16
Baumker et al. [6]	LI-COR LI-190	Artificial light and solar radiation	99.1		
Leon-Salas et al. [7]	Spectroradiometer BlackComet	Artificial light and solar radiation		6.83	
Larochelle et al. [15]	LI-COR LI-190	Solar radiation	99.9	6.78	4.82

conditions. The method enables for the first time to achieve a competitive quantum response curve of the multi-spectral sensor, thereby demonstrating its suitability for measuring PAR at a performance level comparable to that of commercial devices based on a single photodiode. The PLS model used was able not only to reduce the channel correlation but also to mitigate the unwanted measurements of filter leakage in the NIR spectrum by using the clear and NIR channels. Incorporating two more channels with coefficients, the filter leakage problem was solved more efficiently than the vector-quantization method with k-means clustering, because it does not require a further correction phase [7]. Consequently, additional NIR filtering is no longer required, enabling a more compact hardware design.

4. Conclusion

The aim of this work was to solve unresolved measurement problems of PAR calibration methods for multi-spectral sensors, which are only accurate and reliable under identical light conditions. In contrast, our method enables both quantitative and qualitative calibration, providing accurate PAR values as well as spectral information. The implemented setup and methodology enables the efficient calibration of a large number of sensors, ensuring measurement repeatability and robustness against environmental influences. The calibration setup consists of a monochromator as a tuneable light source, a reference sensor, and the DUT. The newly tailored calibration framework begins with the data acquisition, followed by a preprocessing step that considers the spectral characteristic of each device in the setup. The PLS calibration model has been optimized for this work to effectively compensate for the spectral leakage of channels in the NIR. This model, by providing one coefficient per channel, permits an efficient calculation of PAR for multi-spectral sensors. The method was tested on four sensors.

Building on the work of [6,7,11], who demonstrated the general feasibility of determining PAR using multi-spectral sensors, this study presents the first demonstration of the quantum response curve for the AS7341 sensor. The four sensors tested exhibit strong alignment

with the ideal quantum response curve, achieving an R² of 94.788 % (± 0.859 % SD), indicating competitive performance. During a field validation experiment, the measurement results show a MAPE between the multi-spectral and reference sensor of 7.36 % on the overcast day, while on the clear day the MAPE decreased to 2.43 %. These results demonstrated improved calibration accuracy comparable to previous studies, using a controlled setup to ensure independence from varying environmental light conditions.

The additional outcome of this work is a dataset comprising multi-spectral sensor response paired with power measurements for four sensors based on the AS7341 chip [21]. This data set is intended as a basis for the scientific community to develop alternative calibration methodologies. Furthermore, it gives the possibility of improving the calibration in the future by analyzing the potential of alternative algorithms.

Outside in the field, the multi-spectral sensors will undergo considerable exposure to UV radiation, humidity, and temperature fluctuations. Further analysis is required to assess the long-term performance of the calibrated sensors in practical settings within a living lab, such as the ECOSENSE forest, and we see this as a natural next step for future work. However, for the specific purpose of validating the calibration and demonstrating robustness under diverse spectral and illumination conditions, we are confident that the current dataset is appropriate.

The developed calibration methodology marks a substantial step toward standardized, spectrum-independent calibration methods for multi-spectral sensors in environmental monitoring. By enabling accurate and reproducible sensor calibration under controlled conditions, the presented setup empowers researchers and engineers to leverage the potential of multi-spectral sensors to outperform or replace current commercial PAR sensors. In addition, the calibration setup developed in this work provides a robust framework that can be transferred to other fields. It can be extended to additional radiometric parameters such as total irradiance and spectrum-weighted quantities, thereby facilitating its application across various multi-spectral sensor types and spectral ranges.

CRedit authorship contribution statement

Johannes Klueppel: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Megan Jurczak:** Writing – original draft, Visualization, Software, Methodology, Investigation. **Ulrike Wallrabe:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Laura Maria Comella:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

There are no conflicts to declare.

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Data availability

The data is available on Zenodo at <https://doi.org/10.5281/zenodo.15479043>.

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